ASSIGNMENT 2

Deadline: 11:59 PM on February 22, 2026

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1 Compulsory Questions

Question 1 (15 %) (Compulsory)

Consider positive and negative rays as $\mathcal{H}$, which is equivalent to the perceptron hypothesis set in 1D. This hypothesis set is often referred to as a "decision stump" due to the shape of its hypotheses.

Specifically, A **positive ray** is defined by a threshold a, where all points $x \geq a$ are labeled $+1$ and all points $x < a$ are labeled -1. Similarly, a **negative ray** is defined by a threshold a, where all points $x \leq a$ are labeled $+1$ and all points $x > a$ are labeled -1. Thus, the hypothesis set $\mathcal{H}$ consists of all possible positive and negative rays defined by different thresholds a.

1. (5%) What is the growth function $m_{\mathcal{H}}(N)$?
2. (10%) What is the minimum break point for $\mathcal{H}$?

Question 2 (20 %) (Compulsory)

Using the VC generalization bound, we know that with probability at least $1-\delta$, the following inequality holds for all $h \in \mathcal{H}$:

$$E_{\text{out}}(h) \leq E_{\text{in}}(h) + \sqrt{\frac{8 \ln\left(\frac{4m_{\mathcal{H}}(2N)}{\delta}\right)}{N}},$$

where $m_{\mathcal{H}}(N)$ is the growth function of $\mathcal{H}$.

1. (10%) Using the Sauer-Shelah lemma, show that the growth function $m_H(2N)$ can be bounded as:

$$m_H(2N) \leq \left(\frac{2eN}{d}\right)^d.$$

   Substitute this bound into the generalization bound to express $E_{\text{out}}(h)$ in terms of $E_{in}(h)$, N, d, and δ.

   Hint: The Sauer-Shelah lemma states that for a hypothesis set H with VC dimension d, the growth function $m_H(N)$ is bounded by:

$$m_H(N) \leq \left(\frac{eN}{d}\right)^d.$$

   The same result can be found at Week 6 slide page 8.

2. (10%) Suppose H has a VC dimension $d = 10$, and you want the generalization error $E_{\text{out}}(h) - E_{\text{in}}(h) \leq 0.1$ with a confidence of 95% ($\delta = 0.05$). Calculate the minimum number of training samples N required to achieve this.

   Hint: You can use online tool like WolframAlpha (https://www.wolframalpha.com/) to sovle N.

Question 3 (15 %) (Compulsory)

Let $\mathcal{H}_{k\text{-int}}$ be the hypothesis set consisting of the union of **at most** k (closed) intervals on the real line. That is, for parameters $a_1 \leq b_1, \ldots, a_k \leq b_k$, define

$$h(x) = \begin{cases} +1, & x \in \bigcup_{j=1}^{k} [a_j, b_j], \\ -1, & \text{otherwise.} \end{cases}$$

(You may allow some of the intervals to be empty, so this class contains all cases with fewer than k intervals as well.)

1. (10%) Show that $\mathcal{H}_{k\text{-int}}$ can shatter $2k$ points on the real line. Provide an argument using the idea of "blocks" of $+1$ labels.

2. (5%) Show that $\mathcal{H}_{k\text{-int}}$ cannot shatter $2k+1$ points. (Hint: consider an alternating labeling $+, -, +, -, \ldots, +$ on $2k+1$ ordered points.) Conclude that the VC dimension $VC(\mathcal{H}_{k\text{-int}})$ is $2k$.

Question 4 (50 %) (Compulsory)

Consider the following simplified learning scenario: assume that the input dimension is one, and the input variable x is uniformly distributed in the interval $[-1, 1]$. The data set consists of 2 points $\{x_1, x_2\}$ and assume that the target function is $f(x) = x^3$. Thus, the full data set is $\mathcal{D} = \{(x_1, x_1^3), (x_2, x_2^3)\}$. The learning algorithm returns the line fitting these two points as g ($\mathcal{H}$ consists of functions of the form $h(x) = ax + b$). We are interested in the test performance (E_{out}) of our learning system with respect to the squared error measure, the bias and the var.

1. (10%) Give the analytic expression for the average function $\bar{g}(x)$.

2. (15%) Write a Python program to simulate the experiment and determine (numerically) $\bar{g}(x)$, E_{out}, bias, and var:

   - Generate $K = 10000$ training sets (each by sampling x_1 and x_2 uniformly from $[-1, 1]$).
   - For each training set $\mathcal{D}_k$, compute $g^{\mathcal{D}_k}(x)$, and numerical squared error measure $E_{out}(g^{\mathcal{D}_k}) = E_x[(g^{\mathcal{D}_k}(x) - f(x))^2]$.
   - Obtain a numerical estimate of $\bar{g}(x)$ as $\bar{g}(x) = \frac{1}{K}\Sigma_{k=1}^{K} g^{\mathcal{D}_k}(x)$.
   - Compute the averaged squared error over all training sets: $E_{out} = \frac{1}{K}\Sigma_{k=1}^{K} E_{out}(g^{\mathcal{D}_k})$
   - Compute at the bias function about x: $bias(x) = (\bar{g}(x) - f(x))^2$, and the variance function: $var(x) = E_{\mathcal{D}}[(g^{\mathcal{D}}(x) - \bar{g}(x))^2]$.
   - Finally, average over x (using the uniform density) to get numerical estimates of bias and variance.
   - Print out the numerical values: E_{out}, bias and var. Produce a plot showing $f(x) = x^3$ along with the average hypothesis $\bar{g}(x)$.

   Attach your code, a screenshot of the print output and the plot in your answer sheet. Then submit an individual code file `q5.py` along with your PDF answer to Blackboard. You may use libraries like `numpy`, `pandas` and `matplotlib` for numerical computations and plotting.

3. (15%) Compare E_{out} with bias+var from your program output. Then compute analytically what E_{out}, bias, and var should be.

Hint: For a uniformly distributed x in $[-1, 1]$, $E[x] = 0, E[x^2] = \frac{1}{3}, E[x^3] = 0, E[x^4] = \frac{1}{5}, E[x^5] = 0, E[x^6] = \frac{1}{7}$.

2 Optional Questions

Question 5 (Optional)

Consider the hypothesis set $\mathcal{H}$ consisting of **axis-aligned rectangles** in 2D. Each hypothesis in $\mathcal{H}$ is a rectangle defined by four parameters: $x_{\min}$, $x_{\max}$, $y_{\min}$, and $y_{\max}$. A point (x, y) is labeled $+1$ if

$$x_{\min} \leq x \leq x_{\max} \quad \text{and} \quad y_{\min} \leq y \leq y_{\max},$$

and -1 otherwise. The illustration is as follows:

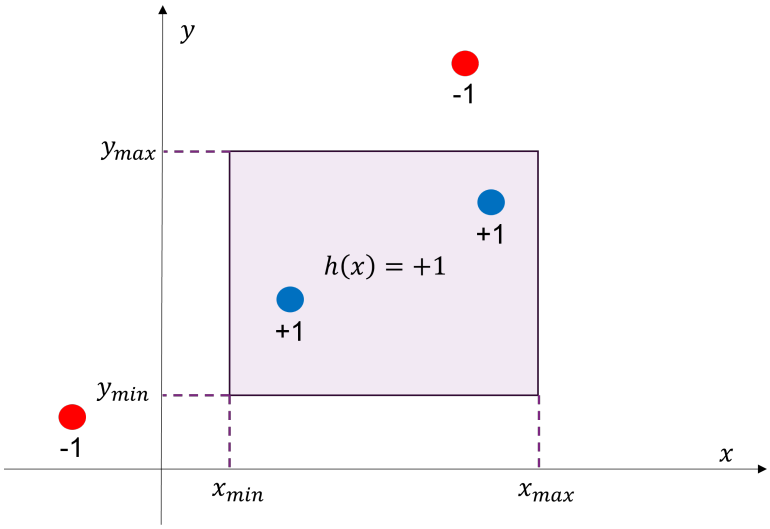

Figure 1: Illustration of an axis-aligned rectangle.

Answer the following:

1. (5%) Show an example of 4 points in the plane that are shattered by $\mathcal{H}$. Draw a figure and provide the x–y coordinates of each point. (You do not need to list all dichotomies.)
2. (5%) Can $\mathcal{H}$ shatter any 5 points? What is the VC dimension of $\mathcal{H}$? Explain your answer. (A formal proof is not required.)

Question 6 (Optional)

For a hypothesis set $\mathcal{H}$ with $VC(\mathcal{H}) = d$, the growth function is bounded by:

$$m_{\mathcal{H}}(N) \leq \sum_{i=0}^{d} \binom{N}{i}.$$

for $N \geq d$.

1. (10%) Derive the expression for the growth function $m_{\mathcal{H}}(N)$ when $N = 3$ and $d = 2$. Show all steps.
2. (10%) Prove that the growth function $m_{\mathcal{H}}(N)$ is polynomial in N when $VC(\mathcal{H}) = d$ is finite. Use the binomial theorem and properties of combinatorial sums.

Hint: The binomial coefficient $\binom{N}{i}$ can be written as: $\binom{N}{i} = \frac{N(N-1)(N-2)\ldots(N-i+1)}{i!}$. $\sum_{i=0}^{d} \frac{1}{i!}$ is a constant when d is finite (it converges to e as $d \to \infty$).

Question 7 (Optional)

Consider the hypothesis set $\mathcal{H}_{\text{2-int}}$ consisting of the union of **at most two** (closed) intervals on the real line. That is, for parameters $a_1 \leq b_1$ and $a_2 \leq b_2$, define

$$h(x) = \begin{cases} +1, & x \in [a_1, b_1] \ \cup \ [a_2, b_2], \\ -1, & \text{otherwise.} \end{cases}$$

(You may allow one of the intervals to be empty, so this class contains the single-interval case.)

1. (4%) Show that $\mathcal{H}_{\text{2-int}}$ can shatter 4 points on the real line. Provide explicit coordinates for 4 points $x_1 < x_2 < x_3 < x_4$ and explain why all 2^4 labelings can be realized.
2. (4%) Show that $\mathcal{H}_{\text{2-int}}$ cannot shatter 5 points. (Hint: consider the labeling $+, -, +, -, +$ on $x_1 < x_2 < x_3 < x_4 < x_5$.)
3. (2%) Conclude the VC dimension $VC(\mathcal{H}_{\text{2-int}})$ and give an explicit upper bound on the growth function $m_{\mathcal{H}_{\text{2-int}}}(N)$ using the standard VC growth bound $m_{\mathcal{H}}(N) \leq \sum_{i=0}^{d} \binom{N}{i}$ when $VC(\mathcal{H}) = d$.

Question 8 (Optional, Programming)

In this question, you will empirically study the generalization behaviour of a simple hypothesis set and compare it with the VC-dimension-based generalization bound from Question 2.

Consider binary classification on $\mathbb{R}$ with inputs $x \in [-1, 1]$ and labels $y \in \{-1, +1\}$ generated by the target function

$$f(x) = \text{sign}(x),$$

with x drawn i.i.d. from the uniform distribution on $[-1, 1]$. Let $\mathcal{H}$ be the hypothesis set of **decision stumps** (positive and negative rays as in Question 1): each $h \in \mathcal{H}$ is specified by a threshold a and a polarity $s \in \{-1, +1\}$ and predicts

$$h(x) = \begin{cases} +1, & \text{if } s \cdot (x - a) \geq 0, \\ -1, & \text{otherwise.} \end{cases}$$

For each training set, you will choose $h \in \mathcal{H}$ that minimises the in-sample classification error $E_{\text{in}}(h)$ on that set (i.e. empirical risk minimisation over decision stumps).

1. (5%) **Implementation.** Write a Python program that, for a given sample size N and number of repetitions K:
   - Repeats the following K times:
     - Draw N i.i.d. samples $x_1, \ldots, x_N$ uniformly from $[-1, 1]$ and label them by $y_i = f(x_i)$.
     - Over a suitable set of thresholds a (e.g. midpoints between sorted x_i's) and both polarities $s \in \{-1, +1\}$, find the decision stump h that minimises $E_{\text{in}}(h)$ on this sample.
     - Estimate the out-of-sample error $E_{\text{out}}(h)$ by drawing a large independent test set (for example, 10^4 new points from $[-1, 1]$) and computing the fraction of misclassified points.
   - Records $E_{\text{in}}(h)$ and $E_{\text{out}}(h)$ for each repetition.

   Your code should print the average in-sample error $\overline{E_{\text{in}}}$ and average out-of-sample error $\overline{E_{\text{out}}}$ over the K runs.

   You may use libraries such as `numpy` for numerical computations.
2. (5%) **Experiments for different N.** Run your program for

   $$N \in \{5, 10, 20, 50, 100, 200\}$$

   with K sufficiently large (for example, $K = 1000$) to get stable estimates. For each N, report the empirical $\overline{E_{\text{in}}}$ and $\overline{E_{\text{out}}}$. Plot $\overline{E_{\text{out}}}$ as a function of N (for example, on a log-scale for N on the x-axis).

   Attach your code, a table of the numerical results, and the plot in your answer sheet.
3. (10%) **Comparison with the VC generalization bound.** Recall that for decision stumps in 1D, the VC dimension is $d = 2$. Using the VC-based generalization bound from Question 2,

   $$E_{\text{out}}(h) \leq E_{\text{in}}(h) + \sqrt{\frac{8 \ln\left(\frac{4 m_{\mathcal{H}}(2N)}{\delta}\right)}{N}},$$

   with the growth function bound $m_{\mathcal{H}}(2N) \leq \left(\frac{2eN}{d}\right)^d$ and a chosen confidence level (e.g. $\delta = 0.05$), compute the theoretical upper bound on $E_{\text{out}}(h)$ for each N using your empirical $\overline{E_{\text{in}}}$.
   - Report these theoretical upper bounds alongside your empirical $\overline{E_{\text{out}}}$ for each N.
   - Briefly discuss:

- How loose or tight the bound appears in practice.
- How both the empirical $\overline{E_{\text{out}}}$ and the theoretical bound behave as N increases.
- Why it is reasonable that the bound is conservative compared to the observed errors.

Homework Submission Policy

- **Full Points:** At most 10 points in total if submitted before the deadline.
- **Late Submission:** At most 8 points in total if submitted within 24 hours after the deadline.
- **No Points:** 0 points if submitted more than 24 hours after the deadline.
- You are encouraged to either type your solution in LaTeX or Word, You can also scan or take photos of your handwritten solution and bundle your scans/photos into one single PDF file.
- All written answers for the **question 1 to question 5** should be submitted in **a single PDF file**. Please also submit the **related code for question 5 q5.py** along with your PDF answer.
- Please submit via Blackboard.